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Three-Tier Knowledge Graph RAG for Legal Question Answering

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Abstract. Accurate legal question answering demands navigating hierarchical document structures, following cross-references, and multi-hop reasoning, capabilities that standard Retrieval-Augmented Generation (RAG) systems lack. Vietnamese legal text compounds this challenge through tonal ambiguity, domain-specific abbreviations, and rigid structural hierarchies. This paper presents a three-tier knowledge graph RAG architecture that integrates a domain ontology, a knowledge graph, and a document tree forest. Three retrieval tiers, LLM-guided tree traversal, dual-level semantic search with intent-aware Personalized PageRank, and Reciprocal Rank Fusion with knowledge graph expansion, operate over these representations. Evaluation on 530 expert-annotated questions across enterprise law, traffic safety law, and postgraduate education regulation shows that the proposed system retrieves at least one correct article in the top 10 results for 92.2% of queries, outperforming the strongest baseline by 7.4%. These results hold across all three domains without domain-specific tuning.

Keywords. Knowledge Graph, Legal Question Answering, Retrieval Augmented Generation, Ontology, Vietnamese NLP, Multi-Tier Retrieval, Multi-Domains

1. Introduction

Accurate access to legal information is critical for citizens, businesses, and legal practitioners navigating regulatory compliance [1]. Legal forums and social networks in Vietnam are flooded with questions on labor law [2], traffic penalties, and regulatory compliance, yet answers are often fragmented, contradictory, or cite repealed provisions.

Legal documents exhibit complex hierarchical structures: a single law organizes content into parts, chapters, sections, articles, clauses, and points, where each level carries distinct legal scope [3,4]. They also contain pervasive cross-references: answering "If a member borrows money to meet charter capital and later withdraws it, what are the penalties?" requires chaining Article 47 (member obligations) to Article 16, Clause 5 (prohibited acts) across different chapters. Vietnamese compounds this challenge through tonal ambiguity, domain-specific abbreviations, and a rigid legal hierarchy.

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Standard RAG [5] retrieves by vector similarity [6], an approach effective in many domains [7] but inadequate for legal text: embeddings blur precise legal meanings, flat retrieval ignores document hierarchy, and dense search cannot chain cross-references. LightRAG [8] and GraphRAG [9] model entity relationships but disregard document hierarchy; RAPTOR [10] builds summary trees via recursive clustering but relies on embedding-only retrieval; PageIndex [11] exploits structure but lacks semantic reasoning; legal ontology systems [12,13] formalize domain knowledge but remain disconnected from retrieval pipelines. To address these limitations, a three-tier knowledge graph RAG is proposed that unifies structural navigation, semantic retrieval, and rank fusion: Tier 1 performs LLM-guided tree traversal, Tier 2 applies dual-level semantic retrieval with intent-aware Personalized PageRank, and Tier 3 merges outputs through Reciprocal Rank Fusion with knowledge graph expansion.

This paper makes three contributions. First, a data engineering pipeline is developed that parses Vietnamese legal documents into hierarchical trees, constructs a domain ontology, and builds a knowledge graph through semi-automatic LLM extraction with expert refinement. Second, a three-tier RAG architecture is designed that unifies structural tree traversal, semantic graph retrieval, and reciprocal rank fusion into a single pipeline; ablation shows the fusion mechanism itself, not merely adding retrieval sources, drives the gains. Third, a multi-intent Personalized PageRank algorithm is introduced that adapts edge weights to detected query intent, allowing the graph to emphasize different relation types for penalty, procedural, or definitional questions. Experiments on 530 questions across three domains show that the system reaches 92.2% top-10 hit rate on legal QA, surpassing PageIndex by 7.4% and RAPTOR by 51.2%, and 88.5% on education regulation, exceeding PageIndex by 13.9%.

2. Related Work

From lexical to neural retrieval. Early legal retrieval relied on lexical matching: BM25 [14] and TF-IDF [15] score documents by term overlap but fail when users paraphrase legal concepts. Dense retrieval shifted to learned representations with Sentence-BERT [6] and DPR [16]; subsequent work added self-reflection [17] and hypothetical document generation [18,7]. Domain-adapted models such as LEGAL-BERT [19] capture legal phrasing yet still operate on flat passages. For Vietnamese, PhoBERT [20] and VnCoreNLP [21] provide foundational models. In [22], Ngo et al. combined BM25 with BERT [23] for the ALQAC 2021 competition. A shared limitation persists across these approaches: treating documents as flat collections discards the hierarchical structure critical for legal text.

Knowledge-enhanced retrieval. To recover structural and relational information, researchers turned to knowledge representations. Legal-Onto [12] is an ontology combining the Rela-model [24] with conceptual graphs for Vietnamese Land Law but relies on manual rules that do not scale. GraphRAG [9] partitions entity graphs into communities with LLM summaries, and LightRAG [8] combines vector search with graph traversal. Both enable multi-hop reasoning yet treat documents as bags of entities without preserving hierarchy; in our experiments, LightRAG's entity extraction quality was low (74% of entities appeared only once). RAG [5] grounds generation in retrieved evidence but remains bottlenecked by retrieval quality. Surveys in [25,26] confirm that bridging legal knowledge with scalable retrieval remains open.

Structure-aware hierarchical retrieval. RAPTOR [10] builds summary trees via recursive clustering and LLM summarization, capturing multi-granularity themes; however, it discards original document hierarchy and relies on embedding-only retrieval at query time, achieving only 41.0% top-10 hit rate on our structured legal benchmark. PageIndex [11] takes a different approach by using LLM-guided tree traversal for hierarchical document search, achieving high precision when the model selects the correct branch but missing content outside the traversal path. Crucially, neither system integrates knowledge graphs for cross-document reasoning, leaving multi-hop queries that span different laws or decrees unresolved.

Currently, there are no existing system combines ontological knowledge, graph-based retrieval, and structural tree navigation for Vietnamese legal text. The proposed system in this manuscript integrates all three capabilities through a three-tier architecture with intent-aware reasoning and rank fusion.

3. Three tier Graph RAG for Legal Question Answering

3.1. Problem Formulation

Vietnamese legal documents form a six-level hierarchy (parts, chapters, sections, articles, clauses, points) with pervasive cross-references across documents. Standard passage retrieval treats documents as flat collections and therefore cannot exploit this hierarchy or perform multi-hop reasoning [27,28]. Each document can be modeled as a rooted tree whose nodes correspond to structural levels and whose edges represent parent–child containment. The full corpus of m documents forms a *document forest* $\mathbf{F} = \{T_1, \ldots, T_m\}$ augmented with cross-reference edges linking articles across trees. Every node is assigned a composite identifier encoding its position in the hierarchy, enabling unambiguous cross-tree references.

Given a user query q, the retrieval task is to return a ranked list of the k most relevant articles from $\mathbf{F}$. Article-level granularity is essential because Vietnamese legal practice requires citing specific articles as the minimal citable unit, not arbitrary text passages.

3.2. System Overview and Knowledge Construction

The architecture separates into an offline phase and an online phase (Figure 1). The offline phase builds three knowledge representations from the raw corpus, following knowledge engineering practice [13]. Each document is first parsed into a rooted tree, forming a document forest with cross-reference edges linking articles across documents. A knowledge graph is then constructed via LLM extraction with expert-in-the-loop refinement. Finally, chapter-level and article-level summaries are generated for LLM-guided branch selection during retrieval. The online phase first analyzes the query (intent classification, abbreviation expansion, seed entity extraction), then processes it through three retrieval tiers: LLM-guided tree traversal (Tier 1), dual-level semantic retrieval over the knowledge graph (Tier 2), and fusion with knowledge graph expansion (Tier 3). Corpus statistics are reported in Table 1; implementation details in Section 4.1.

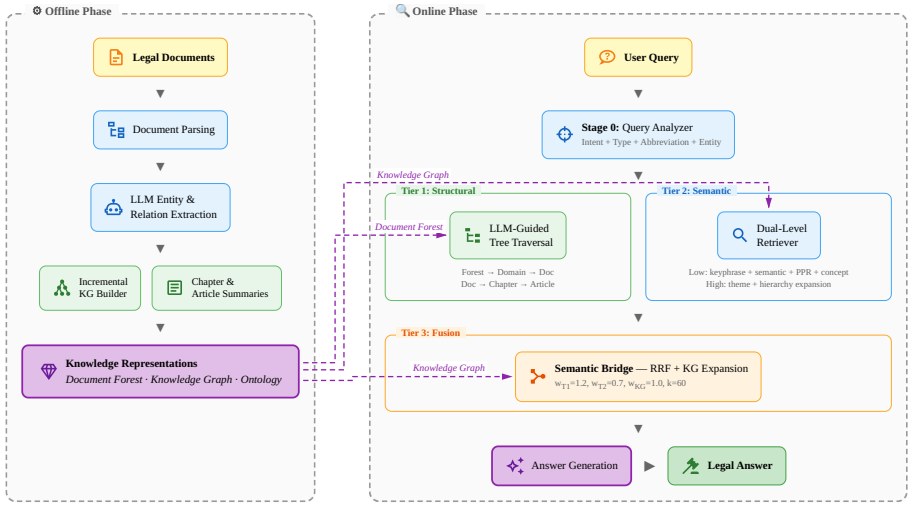


Figure 1. Architecture of the proposed three-tier retrieval system. Offline phase (left): knowledge model construction from legal documents. Online phase (right): query processing through Tier 1 (structural tree traversal), Tier 2 (dual-level semantic retrieval), and Tier 3 (RRF fusion with KG expansion).

3.3. Hybrid Retrieval Architecture

Query Analysis. Given a raw user query q, the query analyzer first classifies its intent into one of six types (Penalty, Definition, Procedure, Requirement, Reference, General) via Vietnamese regex with LLM fallback, then expands abbreviations (e.g., *TNHH* $\rightarrow$ *trach nhiem huu han*) and extracts seed entities such as article references, law references, and legal keywords. These outputs feed directly into the retrieval tiers: Tier 1 uses the expanded query for tree navigation, while Tier 2 uses the seed entities to initialize PageRank and the intent label to adjust edge weights (1.0 for primary relations, 0.5 for others).

Tier 1: LLM-Guided Tree Traversal. Tier 1 takes the expanded query and the document forest, and returns a ranked list of candidate articles with structural scores. It navigates the forest using LLM reasoning in three progressive loops: Loop 0 selects the top 5 documents by matching the query to domain groups via pre-computed keywords; Loop 1 narrows to the top 8 chapters using chapter summaries; Loop 2 identifies the top 7 articles per chapter from article summaries. General Provisions chapters are automatically included. Each loop produces a reasoning trace for interpretability.

Tier 2: Dual-Level Semantic Retrieval. Tier 2 takes the seed entities and intent label from query analysis together with the knowledge graph, and returns a separate ranked list of articles with semantic scores. It operates at two granularities. The *low level* (entity-specific) computes four per-article signals. Keyphrase matching measures syllable n-gram overlap between query terms and KG entity names, aggregated per article as $\max + 0.3 \ln(n+1)\,\bar{s}$ over n matched entities. Semantic similarity computes query–article cosine distance from sentence embeddings. Intent-aware PPR is seeded from entities whose keyphrase score ≥ 0.3. Concept matching propagates keyphrase hits to articles

sharing the same ontology entity types, scoring by type-density. These are combined by weighted sum (w_{kp}=0.05, w_{sem}=0.20, w_{ppr}=0.25, w_{con}=0.20) and normalized to $[0,1]$. The *high level* (conceptual) computes two signals. Theme matching embeds the query against pre-built theme clusters and maps source entities to articles. Hierarchy expansion then finds articles sharing entity types with theme matches (0.5 decay). The two levels are fused: $s(a)=0.70\cdot s_{\text{low}}(a)+0.15\cdot s_{\text{theme}}(a)+0.15\cdot s_{\text{hier}}(a)$, summing to 1.0. Articles below threshold 0.1 are discarded. The PPR component computes query-specific node importance on the knowledge graph [29]:

$$\text{PPR}(v)=(1-\alpha)\sum_{u\in N(v)}\frac{w(u,v)\cdot\text{PPR}(u)}{d(u)}+\alpha\cdot p(v) \tag{1}$$

where $\alpha=0.15$ is the teleport probability, $w(u,v)$ is the edge weight adjusted by query intent, $d(u)$ is the weighted out-degree, and $p(v)$ is the personalization vector seeded from matched entities.

Tier 3: Semantic Bridge (RRF Fusion). Tier 3 produces the final top-k articles through three steps: knowledge graph expansion, rank fusion, and adjacent enrichment. First, seed articles from Tier 1 and Tier 2 are expanded through the knowledge graph's relation edges. Strong relations (*references*, *requires*, *depends_on*) are traversed both cross-chapter (up to 5 articles) and same-chapter (up to 2 articles), with a semantic relevance filter (cosine similarity ≥ 0.4) to suppress noise. This produces a third ranked list alongside Tier 1 and Tier 2. The three lists are then merged using Reciprocal Rank Fusion [30]:

$$\text{RRF}(d)=\sum_{s\in\{T1,T2,KG\}}w_s\cdot\frac{1}{k+\text{rank}_s(d)} \tag{2}$$

where $k=60$ and source weights $w_{T1}=1.2$, $w_{T2}=0.7$, $w_{KG}=1.0$ are empirically set on a development set and shared across all domains. Tree traversal receives the highest weight as it provides the most reliable structural signal. Finally, adjacent articles (± 2) from each fused result are added with decaying scores, as correct answers often lie in neighboring articles. The final ranked list is passed to an LLM for answer generation.

4. Experiments

4.1. Data Collection and Implementation

Legal corpus. 18 Vietnamese legal and regulatory documents span three domains: enterprise law (Law 59/2020/QH14 and nine decrees, 77 chapters, 840 articles), traffic safety law (Law 36/2024/QH15 and two penalty decrees, 195 articles), and education regulation (five postgraduate documents, 109 articles). The corpus totals 949 articles across 77 chapters. Document types range from primary laws enacted by the National Assembly to implementing decrees and university regulations, reflecting the hierarchical nature of the Vietnamese legal system.

Evaluation questions. 530 questions were collected from Vietnamese Facebook legal advice groups (400 legal questions) and UIT student forums (130 education ques-

tions). Questions range from single-article lookups ("What is the minimum charter capital for a securities company?") to multi-hop queries requiring cross-document reasoning. A domain expert labeled each question with 1–8 relevant articles (average 2.3).

Document parsing and summaries. The system is implemented in Python 3.11. Legal texts are parsed with custom regex parsers that detect Vietnamese hierarchy markers ("Chuong", "Muc", "Dieu", "Khoan", "Diem") and build six-level trees. Scanned education PDFs (118 pages) are processed via LLM Vision OCR. Cross-references between documents are extracted via pattern matching on article citations (e.g., "Dieu 47 Luat Doanh nghiep"), producing 168 cross-document edges. Chapter-level and article-level summaries are generated via Claude 3.5 Haiku[2], producing keyword-rich descriptions that Tier 1 uses for branch selection without reading full article text.

Knowledge graph construction. Entity extraction and relation identification use Claude 3.5 Haiku via the Anthropic API, with one LLM call per article. Vietnamese slug normalization removes diacritics and collapses tonal variations to produce canonical entity identifiers. A multi-step entity resolver (exact match, synonym resolution, abbreviation expansion) deduplicates equivalent entities across documents. The domain ontology defines 14 entity types (e.g., Action, Legal Term, Organization, Penalty, Condition) using OWL classes with a 4-level hierarchy and Vietnamese label mappings. The resulting knowledge graph contains 5,936 entities and 5,963 relations across 56 relation types. After automated extraction, a one-time offline expert-in-the-loop step added 267 relations (4.5% of the total) and 14 concept entities, correcting misidentified relations and adding domain-specific links the LLM missed; these curation targets were informed by error analysis on a development subset, and a fully held-out evaluation of the automated-only pipeline is left to future work. This refinement is offline and absent from the online query path.

Online query processing. Tree traversal and article selection use Claude Sonnet 4 via the Anthropic API. Intent classification combines Vietnamese regex patterns for common query structures with LLM fallback for ambiguous cases. Sentence embeddings are computed with paraphrase-multilingual-MiniLM-L12-v2[3] [6], a 384-dimensional multilingual model chosen for its strong cross-lingual transfer to Vietnamese. Vector similarity search uses FAISS[4]. Personalized PageRank is computed iteratively (max 50 iterations, tolerance 10^{-6}, damping α=0.15). All LLM responses are cached in SQLite for reproducibility and cost efficiency.

4.2. Experimental Setup

Metrics. Since each query has expert-annotated relevant articles, we adopt standard retrieval metrics [15] that enable objective, reproducible evaluation against known ground truth, rather than LLM-as-judge scores used in open-ended generation benchmarks. Top-k hit rate measures the fraction of queries for which at least one relevant article appears in the top-k results:

$$\text{Hit@}k = \frac{1}{|Q|}\sum_{q}\mathbf{1}[\exists a \in R_q : \text{rank}(a) \leq k] \tag{3}$$

[2] https://www.anthropic.com/claude

[3] https://huggingface.co/sentence-transformers/paraphrase-multilingual-MiniLM-L12-v2

[4] https://github.com/facebookresearch/faiss

where R_q denotes the set of relevant articles for query q. Mean Reciprocal Rank averages the reciprocal rank of the first relevant result:

$$\text{MRR} = \frac{1}{|Q|} \sum_{q} \left(\min_{a \in R_q} \text{rank}(a) \right)^{-1} \tag{4}$$

. Results are reported at $k \in \{1, 3, 5, 10, 20, 30\}$, with top-10 hit rate and MRR as the primary metrics. Top-10 hit rate is prioritized because in a RAG pipeline the LLM validates retrieved context, so one correct article among ten candidates suffices. MRR complements it by rewarding higher placement of relevant articles. End-to-end answer quality is not separately scored: with the answer LLM and IRAC generation prompt held fixed, the retrieved context is the only variable across methods, so answer quality tracks retrieval quality; scoring the implemented IRAC generator is left to future work.

Table 1 summarizes the constructed knowledge graph.

Table 1. Knowledge graph statistics.

Entity Type	Count	Relation Type	Count	Structure	Count
Total Entities	5,936	Total Relations	5,963	Tree Nodes	9,226
Action	1,903	requires	1,377	Documents	13
Legal Term	1,632	applies_to	810	Chapters	77
Organization	500	has_authority	506	Articles	840
Legal Reference	451	references	434	Cross-refs	168
Other (9 types)	1,450	Other (52 types)	2,506		

Baselines. Traditional: BM25 and TF-IDF with syllable-level tokenization. RAG systems: NaiveRAG (dense retrieval, no reranking); LightRAG (hybrid KG retrieval); RAPTOR (collapsed-tree with LLM summaries); PageIndex (two-step hierarchical tree search). All systems use Claude Sonnet 4 for LLM calls, retrieve from the same full corpus, and use their authors' released default hyperparameters for a fair comparison.

4.3. Results

Table 2. Legal domain hit rate (%) on 400 questions.

Method	Top–5	Top–10	Top–20	Top–30	MRR
Traditional retrieval					
BM25	30.8	47.5	66.2	74.8	0.200
TF-IDF	29.2	45.8	64.5	74.8	0.207
Keyword	33.5	48.5	63.2	72.8	0.222
RAG systems					
NaiveRAG [6]	22.8	32.8	51.8	60.5	0.146
LightRAG [8]	25.5	39.8	47.8	49.2	0.173
RAPTOR [10]	25.2	41.0	54.5	62.3	0.170
PageIndex [11]	**76.2**	<u>84.8</u>	<u>90.2</u>	<u>91.0</u>	<u>0.559</u>
3-Tier (ours)	<u>74.5</u>	**92.2**	**94.8**	**95.0**	**0.603**

Bold is the best, <u>underlined</u> is the second.

As Table 2, the proposed system achieves $92.2 \pm 0.3\%$ top-10 hit rate (MRR 0.603), outperforming PageIndex and BM25. The gain over PageIndex, though modest, is statistically significant (McNemar's exact test $p = 1.8 \times 10^{-4}$; bootstrap 95% CI of the gap $[+3.8, +11.5]$ pp), and the margins over all RAG baselines exceed $+51$ pp ($p < 10^{-54}$). PageIndex benefits from tree navigation but misses cross-reference articles outside the traversal path. RAG baselines (NaiveRAG 32.8%, LightRAG 39.8%, RAPTOR 41.0%) underperform BM25 (47.5%), that means general-purpose embeddings lack Vietnamese legal vocabulary, generic KG extraction introduces noise, and RAPTOR's summaries lose article-level granularity. Our system overcomes these limitations via LLM-guided navigation and domain-specific knowledge graph.

Table 3 reports per-domain results (education as a generalization test). The system outperforms all baselines in every domain: traffic law scores highest (98.0%, direct violation-to-penalty mappings), enterprise law is harder (86.5%, cross-decree references), and education (88.5%) shows the largest margin over PageIndex (+13.9%). The same pipeline and hyperparameters were used across all domains without per-domain tuning.

Table 3. Per-domain top-10 hit rate (%) across three domains.

Method	Enterprise (197)	Traffic (203)	Education (130)
NaiveRAG	29.9	35.5	50.0
LightRAG	20.3	58.6	53.1
RAPTOR	18.5	63.5	53.8
BM25	31.5	63.1	70.0
TF-IDF	39.1	52.2	68.5
Keyword	20.3	75.9	56.2
PageIndex	76.6	92.6	74.6
3-Tier (ours)	**86.5**	**98.0**	**88.5**
Ours vs. PageIndex	*+9.9*	*+5.4*	*+13.9*

4.4. Ablation study

PageIndex has a slightly higher top-5 hit rate (76.2% vs. 74.5%): KG expansion occasionally displaces tree results from positions 4–5 to 6–7 but recovers cross-reference content PageIndex never reaches (a 17.7 pp gain from positions 6–10 vs. 8.6 pp). In a RAG pipeline filtering ten candidates, this tradeoff is favorable.

Per-category top-10 hit rate ranges from 0% (household business, outside the corpus) to 100% (speed penalties). Of the 31 missed queries (7.8%), 27 involve enterprise law (cross-decree dependencies, out-of-corpus articles, and $>$2-article queries), mostly linked by implicit reasoning chains rather than explicit cross-references.

Ablation. Table 4 isolates each component. A naive merge of Tier 1 and Tier 2 without RRF falls *below* Tier 1 alone (unfused Tier 2 injects noise into the top 10), so gains stem from fusion, not added sources. Ablating intent-aware PPR costs +1.0 pp (legal) and +4.6 pp (education): its effect is modest on legal, where tree traversal already resolves most queries, but larger on education, where semantic routing matters more.

Table 4. Component ablation (top-10 hit rate, %).

Configuration	Legal (400)	Education (130)
Tier 2 only (KG semantic)	40.5	50.8
Tier 1 only (tree traversal)	75.8	79.2
Tier 1 + Tier 2, no RRF	54.0	60.8
Full, no intent-PPR	91.2	83.9
Full (3-tier + RRF)	**92.2**	**88.5**

5. Conclusion and Future work

This paper presented a three-tier knowledge graph RAG for Vietnamese legal QA, unifying structural tree traversal, semantic graph search with intent-aware Personalized PageRank, and Reciprocal Rank Fusion. It achieves 92.2% top-10 hit rate on 400 legal questions (+7.4 pp over PageIndex, +51.2 pp over RAPTOR) and 88.5% on 130 education questions (+13.9 pp), using the same pipeline across all three domains without per-domain tuning.

Current limitations include Tier 1 latency (around 7 s per query) and weaker performance on enterprise queries needing implicit multi-document reasoning. Future work will investigate parallel LLM calls and distillation for latency, automated reasoning-chain discovery, and extension to more regulatory domains.

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